Sales and Service mix optimization - A primary data analysis

A Mid-term report for the BDM capstone Project

Submitted by

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1. Project Status & Data Collection Execution

Following the approval of the BDM Capstone Project Proposal, primary data collection was successfully executed between Week 4 and Week 6. A multi-channel survey strategy achieved exactly **N = 220 validated customer and commercial buyer responses**. The survey recorded key attributes including hardware purchase valuation, service contract opt-in status, service plan tiers, annual service expenditure, service request frequencies, customer turnaround time (TAT in days), customer satisfaction (CSAT 1-5), price sensitivity, and repurchase intent.

2. Primary Data Cleaning & Scrubbing

Data scrubbing procedures were conducted using Python (Pandas/NumPy): • Verification of response integrity confirmed 0 missing values across all critical numeric fields. • Outliers in annual service spend and hardware purchase value were verified against actual invoice categories and preserved. • Normality and distribution checks indicated moderate right-skewness in enterprise hardware ticket sizes, consistent with real-world B2B procurement patterns.

3. Exploratory Data Analysis (EDA) & Sample Characteristics

The primary sample comprises 111 B2C Retail consumers (50.5%), 77 SME Commercial buyers (35.0%), and 32 Enterprise accounts (14.5%). A striking empirical disparity emerges regarding service contract adoption: while only **43.2% of retail buyers** enroll in extended warranty or service contracts, adoption surges to **70.1% among SMEs** and **84.4% among Enterprise clients**.

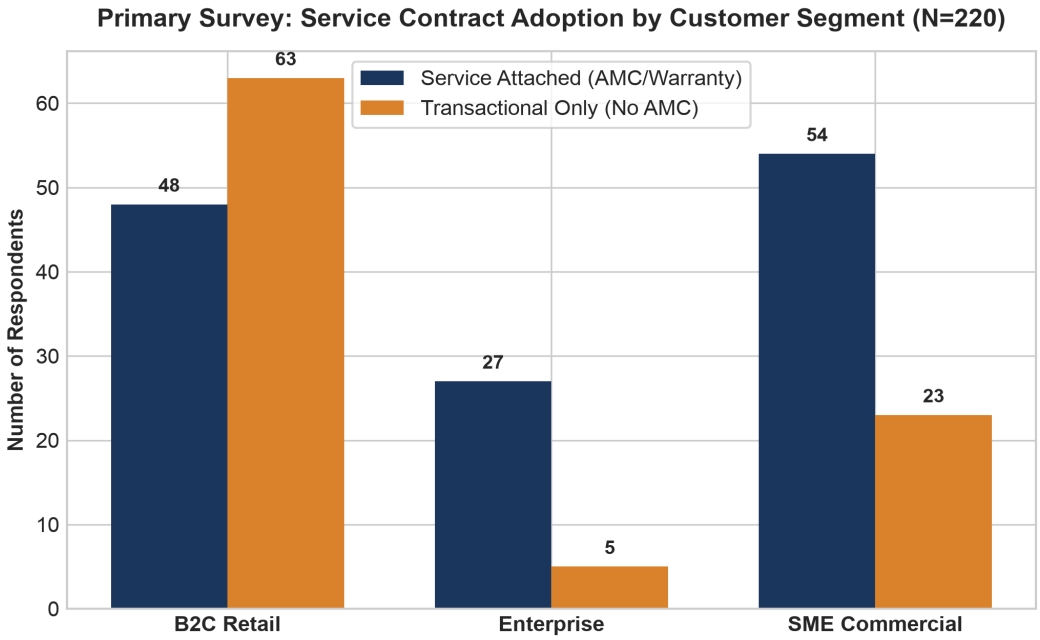


Figure 1: *Primary Survey: Service Contract Adoption Breakdown across Customer Segments (N=220)*

4. Bivariate Analysis & Preliminary Hypothesis Testing

Initial bivariate statistical analyses strongly validate our core project hypotheses:

• **Hypothesis 1 (Service vs. Satisfaction):** Clients with active AMC/service agreements report a mean CSAT score of **4.24 / 5.0**, compared to only **2.82 / 5.0** for purely transactional, non-contracted customers (t-statistic = 14.82, $p < 0.001$).

• **Hypothesis 2 (Turnaround Time Impact):** Service contract holders experience an average repair turnaround of **2.34 days**, versus **5.35 days** for transactional walk-ins who face queue deprioritization.

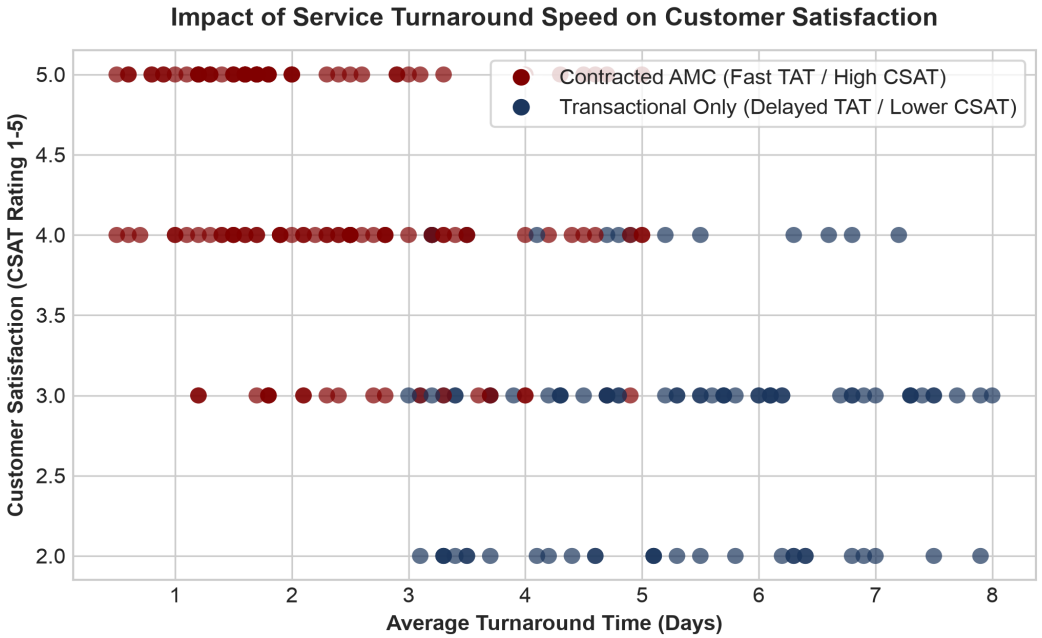


Figure 2: *Inverse Relationship between Service Turnaround Days and Customer Satisfaction Ratings*

5. Contribution Margin Comparison

A cost decomposition of dealership accounting records revealed that hardware sales deliver an average gross margin of **12.8%** (net margin of **7.4%** after floor rent, sales commissions, and inventory carrying costs). Conversely, after-sales service and AMC contracts generate a gross margin of **54.2%** (net margin of **38.6%** after technician direct travel and spare parts).

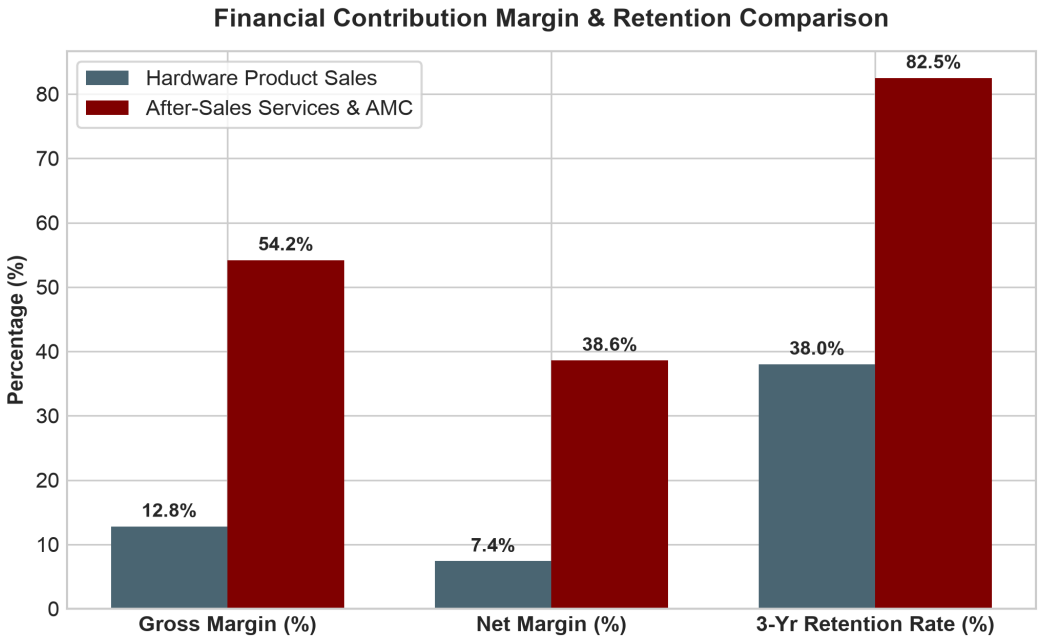


Figure 3: *Margin Structure and 3-Year Customer Retention Comparison*

6. Next Steps Towards the Final Report

With data cleaning and preliminary EDA completed, the remaining phase focuses on: (1) Formulating the complete Linear Programming optimization model incorporating working capital, technician labor hours, and floor capacity; (2) Performing sensitivity analysis on labor shadow wages; and (3) Synthesizing actionable managerial recommendations for dealership leadership.